THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of 4-digit palindromes with first digit 5 is: 1

(a) 9 **(b)** 100 **(c)** 10 **(d)** 90

2. The sum of 8 consecutive counting numbers starting from 25 is: 1

(a) 228 **(b)** 236 **(c)** 200 **(d)** 264

3. A 5-digit number ends with digit 4 and has digit sum 28; the sum of its other four digits is: 1

(a) 24 **(b)** 33 **(c)** 32 **(d)** 15

4. The number of digits in the largest palindrome smaller than 1000 is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

5. On reversing and adding 265, the first sum is: 1

(a) 816 **(b)** 827 **(c)** 838 **(d)** 926

6. 59 is added to its reverse and the sum is 154. How many reverse-and-add steps in all are needed to reach a palindrome? 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 2

7. How many steps does the Kaprekar routine take to reach 6174 from 7418? 1

(a) 6 **(b)** 5 **(c)** 8 **(d)** 7

8. The hailstone (Collatz) sequence of 15 has how many terms (including 15 and 1)? 1

(a) 17 **(b)** 18 **(c)** 16 **(d)** 19

9. **Assertion (A):** A Kaprekar step on 5432 gives 5432 - 2345 = 3087. 1
Reason (R): A Kaprekar step adds the largest and the smallest arrangements of the digits.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The smallest 3-digit number that can be formed from the digits 4, 0, 6 is 046. 1
Reason (R): A 3-digit number cannot have 0 in the hundreds place.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the number of terms in the hailstone (Collatz) sequence of 18 and its largest term. 2

12. A number has 14 digits, all equal to 7. Find its digit sum, and say whether the number is divisible by 3 and whether it is divisible by 9. 2

13. How many steps does the Kaprekar routine need for 4651? Write the chain. 2

14. Find the number of terms in the hailstone (Collatz) sequence of 26 and its largest term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find a 4-digit palindrome of the form $abba$ with $a = 3$ and $b = 6$. Show that it is divisible by 11. 3

16. A row of 7 numbers starts 3, 5 and each next number is the sum of the previous two. Find the 7th number and the sum of all 7 numbers. 3

17. How many 4-digit numbers can be formed from 4, 5, 6, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

OR

Find a 4-digit palindrome of the form $abba$ with $a = 4$ and $b = 9$. Show that it is divisible by 11. 3

18. A row of 8 numbers starts 2, 6 and each next number is the sum of the previous two. Find the 8th number and the sum of all 8 numbers. 3

OR

How many 4-digit numbers can be formed from 1, 2, 8, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Priya plays the hailstone game. If the number is even she halves it; if it is odd she multiplies it by 3 and adds 1. She stops when she reaches 1. She starts with 6. 5

(i) What number comes after 6?
(ii) What number comes after 3?
(iii) How many steps does she take to reach 1 from 6?
(iv) Write the sequence starting from 5, say how many steps it takes and name the largest number reached.

20. A palindromic number of 5 digits is formed as $abcba$ where $a = 4$, $b = 6$, $c = 2$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

OR

Find the number of steps the reverse-and-add process takes from 94 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. 5

END OF QUESTION PAPER — Total: 40 Marks